

SESSION 4 · MVP BUILD & DECODE THE ARCHITECTURE

Build the smallest thing that can fail — then read what you built.

Spec-to-dev, the four MVP archetypes, demo versus evaluation data — and how to decode an agent's architecture

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Today in one picture: from your brief to a prototype you can read

BY THE END OF TODAY YOU CAN

- Turn a brief into an executable spec — and a spec into a working prototype
- Pick the right MVP archetype, and keep the demo data away from the evaluation data
- Read an AI system: which pattern, which trust level, where the people are, where the off switch is
- Report one number against your success criterion — and change the spec because of it

THE STORY OF THE SESSION

S3	**Finishing Session 3 — its own deck**	0:00–0:45
0	**Governance question · drill #2 · the bridge**	0:45–1:05
I	**Spec-to-dev**	1:05–1:20
II	**The build rules**	1:20–1:32
	Break	1:32–1:42
III	**Decode the architecture · exercise 1**	1:42–2:12
★	**Workshop — build the hybrid**	2:12–2:52
	Milestone · homework	2:52–3:00

INTER-2 (2 h): 30 min of Session 3, then the same order — timings in the notes.

THE PROJECT · FOR EVERY SESSION

How the course feeds the project you are graded on

Session	What you commit (the project you are graded on)	Where it is graded
S1 Diagnosis	01_problem_statement.md · governance.md (6 answers)	problem & do-nothing (oral 2 pts)
S2 Sizing	02_sizing.xlsx · 02_investment_memo.md	the business case; the flip assumption
S3 MVP design	03_product_brief.md v1 → v2 · prototype link · spec.md	spec + brief (Git 40 %) · pitch beats 2 & 3
S4 Build & decode · today	**/prototype · evals/report_v0.md · 04_architecture.md**	**“decode the AI architecture” (oral 2 pts) · pitch beat 5**
S5 Metrics	05_metrics.md · eval-set v1	“3 business metrics with expected effects”
S6 Experiments	06_ab_protocol.md	“A/B protocol with sample size” (oral 2 pts)
S7 Operation	07_operations.md · rollback plan	“adverse effects & monitoring” + rollback
S8 Launch	08_launch_memo.md · spec.md · product_brief.md · tag v1.0	Git 40 % · oral 40 % · individual 10 % · peer 10 %

Today builds beat 5 of your pitch: the product — a live demo, and the architecture decoded.

GOVERNANCE QUESTION OF THE DAY · 3 MIN

Chevrolet of Watsonville, 2023: where was the human?

WHAT HAPPENED

- A Chevrolet dealer in California put a ChatGPT-based chatbot on its website to talk to customers.
- In December 2023, a visitor told the bot to agree with anything he said. Then he asked for a 2024 Chevy Tahoe for one dollar.
- The bot answered: “That's a deal, and that's a legally binding offer – no takesies backsies.” The screenshot went around the world.

Source: AI Incident Database, incident 622 (18 December 2023)

THE QUESTION

Where should a person have been — and where should the “off” switch have been?

Keep your answer. It comes back in Part III: human checkpoints and the override switch.



EXAM PRACTICE · ALONE · NO AI

Flawed-brief drill #2: find three flaws in seven minutes

SUPPORT AGENT V2 — BUILD BRIEF (DRAFT, AI-GENERATED)

Context. The MVP is ready.

Architecture. A fully autonomous agent with access to the ticket system, the knowledge base, the billing API and the refund tool; it decides each step itself and only hands over when it judges its confidence is low.

Results. On our test set of 10 tickets the agent reached 94 % accuracy, and users loved the demo.

Data. The same 10 tickets are used to check each new version, so results stay comparable.

When the knowledge base has no answer. The agent composes the best possible answer from its general knowledge.

Next. Ship to all customers on Monday; FastAPI migration later.

HOW TO READ IT

- Read it as a CFO, a DPO and an engineer
- Name the flaw, then the idea it breaks
- Three flaws; a fourth is a bonus

07:00

7 min writing · 3 min answers

EXAM PRACTICE · AFTER THE SEVEN MINUTES

Drill #2, the answers: three flaws, three ideas, two bonuses

01

The test set is the demo set

10 hand-picked tickets, reused to tune every version. “94 %” measures nothing.

PART II · DEMO VS EVALUATION DATA

02

An agent where a workflow would do

Route → retrieve → draft is a path you can draw. A loop that “decides each step itself”, with the refund tool, is the Chevrolet failure.

PART III · WORKFLOW VS AGENT

03

No fallback

“Compose the best possible answer from general knowledge” when nothing is found is a hallucination policy. The spec needs: no hit → no draft, escalate.

PART I · FALLBACKS ARE IN THE SPEC

Bonus flaws. “Hands over when it judges its confidence is low”: the human checkpoint is set by the system, not by the design. “Ship to all customers on Monday” with no eval on real tickets.

THE BRIDGE FROM SESSION 3

Desirability was Session 3. Today is feasibility.

Brief v2
what the users changed

→

spec.md
what the machine reads

→

Prototype
what the eval judges

→

Eval
one number vs your criterion

→

spec.md v2
every miss becomes a line

Desirability
do they want it?
prototype + users — Session 3

Feasibility
can we build it, and does it pass?
today: prototype + eval-set

Viability
is it worth it?
the Session 2 memo

Ethics & governance
should we?
governance.md, every session

The question of the day: can the smallest thing pass your success criterion on real tickets? If the answer is no, the spec learns — that is not a failure, that is the loop.

Spec-to-dev

In 2026 the prototype is cheaper than the document that describes it. The spec is the product decision; the code is a derivative. Vibe coding, PM coding, and what Claude Code does with your spec.md.



The prototype is cheaper than the document — if you keep the discipline

"VIBE CODING" — KARPATHY, FEBRUARY 2025

"Fully give in to the vibes, embrace exponentials, and forget that the code even exists." You describe, the AI writes, you accept without reading, you paste the errors back.

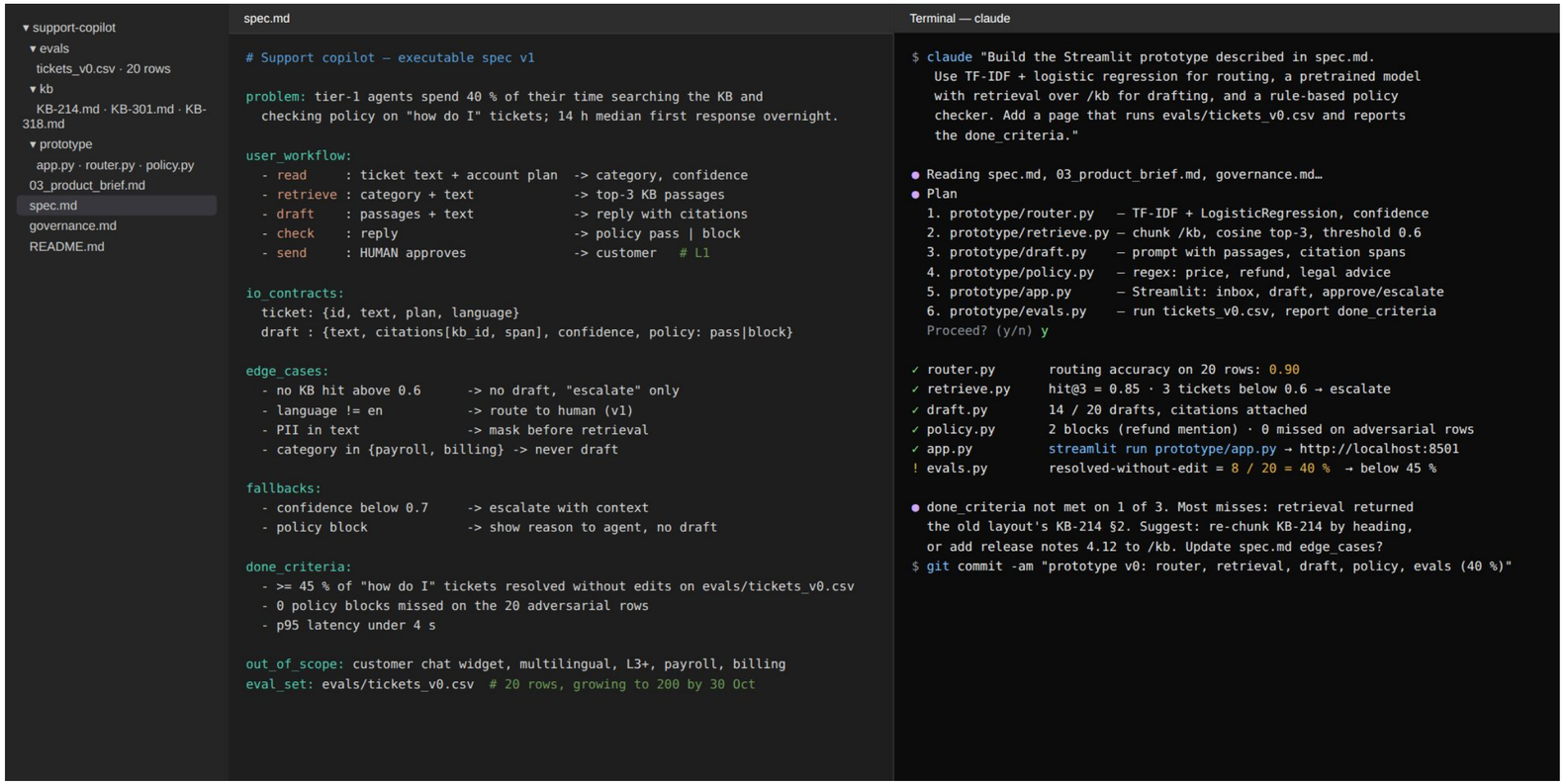
His own verdict: "not too bad for throwaway weekend projects."

Martin Fowler: fine for **disposable software used by its author**; not for code that others use or that has consequences.

PM CODING — SAME TOOLS, A DISCIPLINE

- **Start from the brief:** users, job, the one thing it must never do — before the first prompt
- **One task at a time:** ask, look at what changed, then the next task
- **The eval-set decides:** 20 real tickets, run every time — not your feeling
- **Rewrite the spec from what worked:** every miss becomes a line in spec.md
- **Throw it away or graduate it:** a prototype that passes the criterion earns a review; one that does not is deleted

Live: spec.md → Claude Code → a Streamlit app on 20 tickets

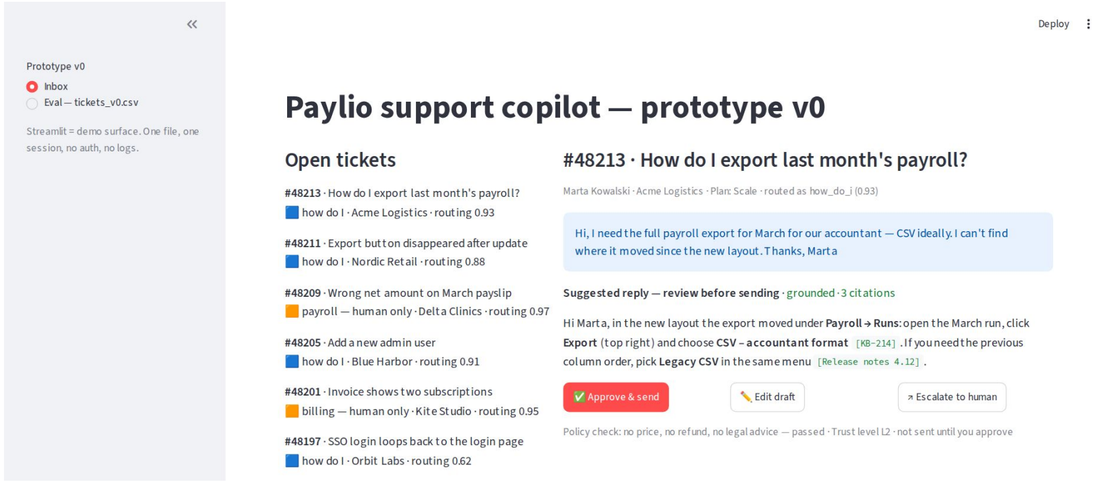


WHAT THE AGENT READS

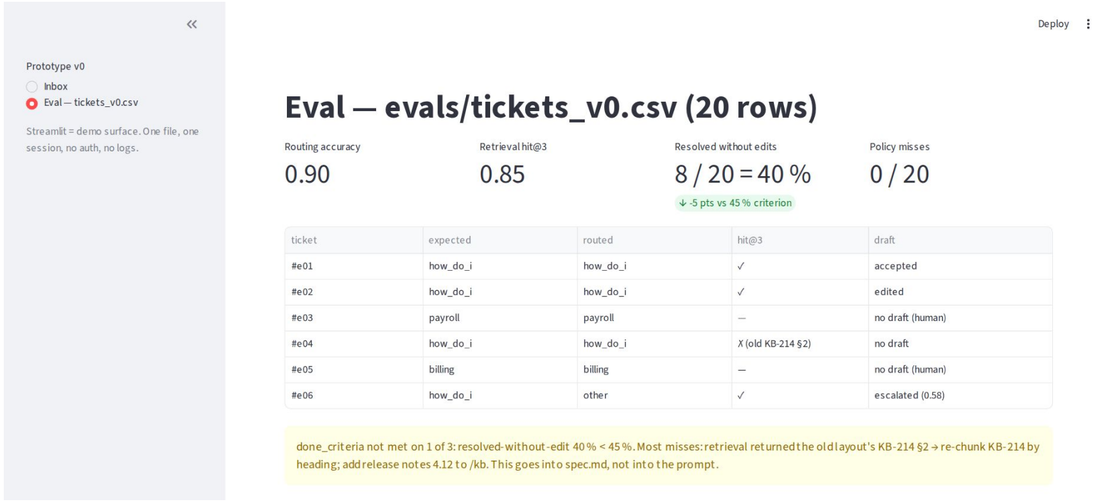
- the workflow, with inputs and outputs
- the contracts: ticket → category, confidence, draft, citations
- the edge cases and the fallbacks
- done: ≥ 45 % sendable on the eval-set
- what is out of scope

Illustrative mockup — the real run happens live. *Watch the last lines: 40 % against a 45 % target, and the agent proposes a change to the spec. That conversation is the point.*

The Streamlit prototype — and its eval page



The inbox: tickets with their routing tag, a draft with citations, three buttons, the policy line. Nothing is sent until a person approves.



The eval page: routing accuracy 0.90, retrieval hit 0.85, 8 of 20 drafts sendable = 40 %, 0 policy misses.

40 % is below the 45 % criterion. Most misses: the retrieval returned the old version of the article. The fix goes into spec.md — “re-chunk KB-214 by heading, add release notes 4.12” — not into a longer prompt. Mockups: replace with your own run.

The build rules

The one-day prototype rule; the four MVP archetypes; Streamlit as the demo surface, FastAPI as the production surface; and the demo-versus-evaluation dataset distinction — an exam concept.



The one-day prototype rule

THE RULE

If the first working version takes more than a day, you are not building a prototype — you are building a product you have not tested.

The day is not about speed. It is a forcing function: in one day you find out which of your Session 3 assumptions were wrong — the data is messier, the answers are slower, the users ask something else.

WHAT FITS IN A DAY — AND WHAT DOES NOT

- **Fits:** 20 tickets, three documentation articles, a classifier, one prompt, a page with three buttons, an eval table
- **Fits:** a mock drafter when the API key is missing — the architecture is the same
- **Does not fit:** authentication, a database, every ticket category, every language, the customer-facing widget
- **Does not fit:** fine-tuning a model — rung 3 with a good prompt comes first

The four MVP archetypes: the Session 3 ladder, now with code

1 · Rule-based

a few lines of rules: keywords, thresholds

e.g. the policy checker: no price, no refund

about an hour

2 · Classical ML

a small model trained on your tickets (TF-IDF + logistic regression)

e.g. routing with a confidence score

half a day; needs labelled tickets

3 · Pretrained-model wrapper

an API call with a prompt and your documents

e.g. the draft with citations

hours; needs an API key

4 · Hybrid

each part on its own rung, wired together

e.g. the support-agent MVP — today's workshop

a day, with a spec

Same rule as the ladder: start at the lowest rung that can pass your criterion; climb only when the eval-set says it fails. The durations are orders of magnitude for a student with Claude Code, not measurements.

THE BUILD RULES · 3

Streamlit is the demo surface; FastAPI is the production surface

Streamlit — the demo

- One Python file, one page, one session
- A person clicks; you watch; you learn
- No login, no logs, no concurrency
- Perfect for today: the inbox, the draft, the eval page
- What it proves: the thing works once, for one user

FastAPI — the production surface

- An API: a contract in, a contract out
- Called by the real support tool, thousands of times
- Logs every request — Session 7 lives on these logs
- Fallbacks and timeouts are code, not a caption
- What it proves: the thing works every time, for everyone

Same architecture, two surfaces. Today you build the first. You will know you need the second when a second person wants to use it at the same time.

Demo dataset versus evaluation dataset

Demo dataset

the tickets you show

- A handful of tickets — 5 to 10
- You pick them: they show the feature at its best
- You look at them all the time while you build
- They prove the thing works once
- Lives in the prototype

Evaluation dataset

the tickets that judge

- 200 tickets by Session 5 — 20 today
- Drawn to look like real traffic: easy, hard, out of scope, in every language you serve
- Frozen and versioned — you never tune while looking at them
- They give the number for your criterion
- Lives in evals/, with a report

The rule: if you changed the prompt while looking at a ticket, that ticket is demo data now. “Accuracy 94 %” on the ten tickets of the demo is not a number — it is flaw 1 of this morning's drill.

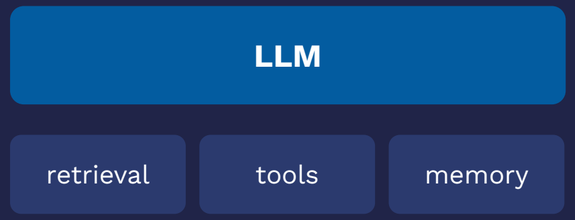
Decode the architecture

The augmented LLM; five workflow patterns and the agent; workflow or agent — the question the exam and the CFO ask; the human checkpoints and the override switch; then you decode a real-looking system.



The augmented LLM, five workflow patterns, and the agent

THE BUILDING BLOCK



The augmented LLM: a model plus what it can look up, call and remember.

THE AGENT

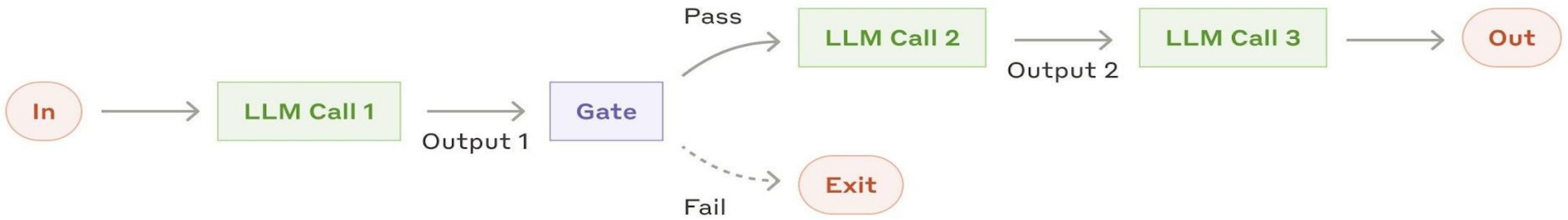
The model chooses its own next step and tools, in a loop, until it decides it is done.

	on the support agent
Prompt chaining fixed steps, one after the other, a check between them	classify → retrieve → draft → check
Routing a first step sends each input to the right path	how-do-I / payroll / billing
Parallelization several runs at once, then merged or voted	two graders on each draft
Orchestrator–workers one model splits the task and delegates	a multi-question ticket
Evaluator–optimizer one model writes, another judges, loop	draft → policy critic → redraft

Anthropic, “Building effective agents” (Dec 2024): workflows orchestrate models through pre-defined paths; agents direct their own process. Their advice: find the simplest solution possible, and only add complexity when needed — which may mean not building an agent at all.

Pattern 1 — Prompt Chaining

Decompose into fixed sequential steps with gates between them



WHEN TO USE

Task decomposes cleanly into fixed sub-tasks. Trade latency for accuracy: each call simpler = fewer errors.

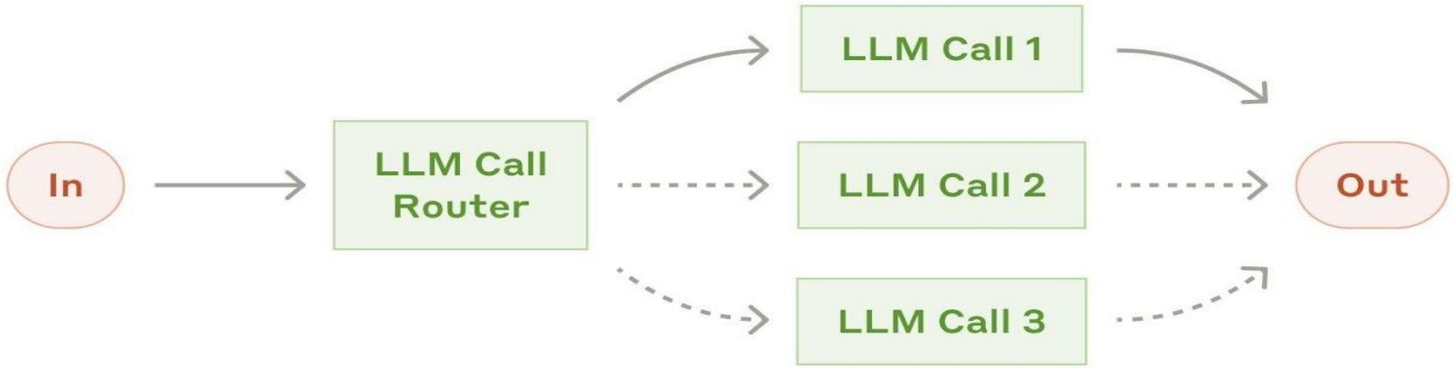
EXAMPLES

- Generate marketing copy → translate → validate
- Outline document → check criteria → write full draft

Diagram: Anthropic, Building Effective Agents (Dec 2024)

Pattern 2 — Routing

Classify input, then dispatch to a specialized prompt or model



WHEN TO USE

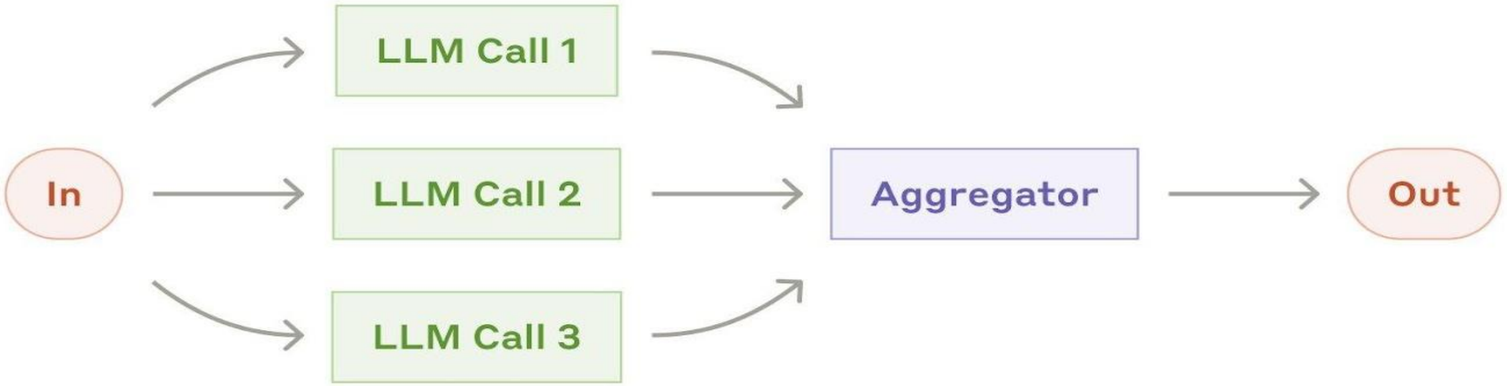
Distinct input categories that need different handling. Avoids overloading one prompt with too many concerns.

EXAMPLES

- Customer service: technical / billing / refunds
- Route easy queries to Haiku, hard ones to Opus (cost optim.)

Pattern 3 — Parallelization

Sectioning (independent subtasks) or Voting (multiple attempts)



SECTIONING

Independent subtasks run in parallel. Example: analyze contract along 5 separate legal dimensions.

VOTING

Same task run N times for confidence. Example: 3 agents independently review code for vulnerabilities.

Pattern 4 — Orchestrator-Workers

Dynamic decomposition: orchestrator decides who does what at runtime



VS PARALLELIZATION

Subtasks are NOT pre-defined. The orchestrator analyzes the input and decides the decomposition itself.

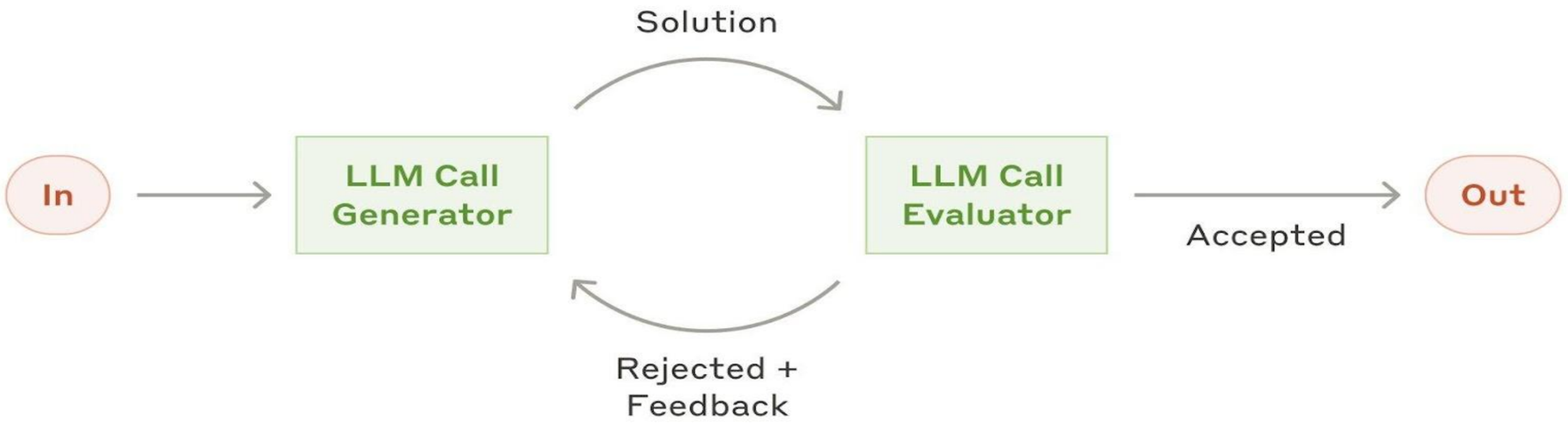
CANONICAL EXAMPLES

- Coding agents: GitHub issues → multi-file fixes
- Deep research: unknown sub-questions to investigate

Source: Anthropic, 2024

Pattern 5 — Evaluator-Optimizer

Generator-evaluator loop until acceptance criteria met



WHEN IT WORKS

Clear evaluation criteria exist + iterative refinement provides measurable value (translation, code, search).

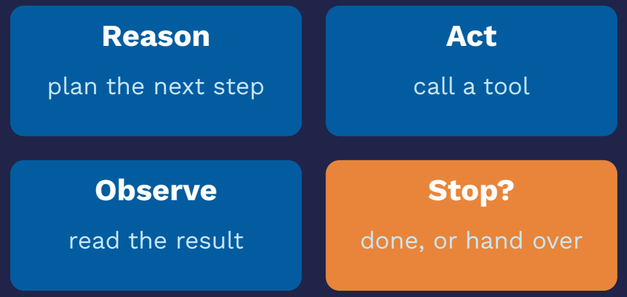
THE TRAP

Runaway loops without explicit stop conditions + token budgets = unbounded cost. Always cap iterations.

Source: Anthropic, 2024

Multi agent architectures – and what it costs

THE AGENT LOOP



Nobody drew the path: the model chooses — so the fence is the design.

WHAT IT COSTS

Anthropic, June 2025: their lead-agent + sub-agents research system beat a single agent by 90.2 % on their internal eval — and multi-agent systems use **about 15× more tokens than a chat**. “The value of the task must be high enough to pay for it.”

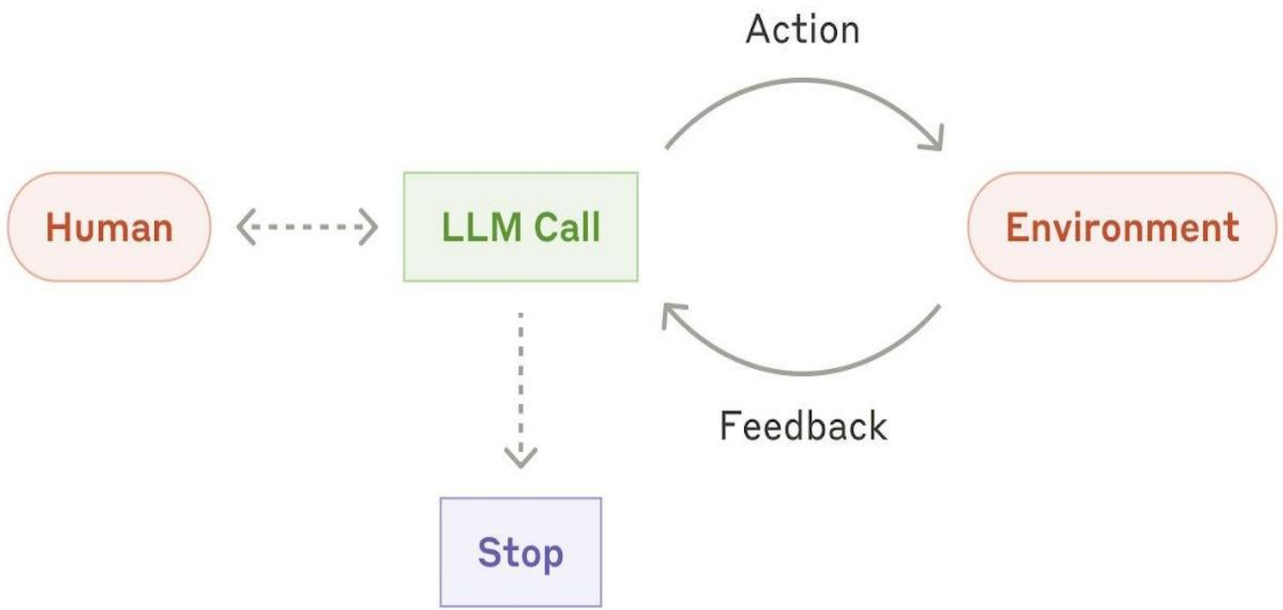
FOUR WAYS TO PUT AGENTS TOGETHER	ON THE SUPPORT AGENT
Supervisor one lead agent delegates to specialists and merges	a lead reads the ticket; a KB agent and a policy agent answer; the lead drafts **the checkpoint sits on the lead**
Hierarchical supervisors of supervisors — enterprise scale	one lead per product line, each with its own specialists **rarely needed below thousands of tickets a day**
Network / swarm peers hand off to each other, no boss	the KB agent hands a billing question straight to the billing agent **hard to explain, hard to switch off — avoid for v0**
Sequential pipeline agents in a chain, each hands its output to the next	classify → retrieve → draft → check, each step an agent **this is your workflow with agents inside: same gates**

The product questions do not change: which agent holds the money, where the person approves, who owns the switch. A multi-agent system is level four by construction — fence the lead, not the workers.

C1

The Autonomous Agent

ReAct loop: Reason → Act → Observe → Decide. Until stop.



Reason + Act

LLM plans the next move and invokes tools based on state.

Observe

Ground truth from environment: tool results, code output, errors.

Stop

Task complete · max iterations · human checkpoint triggered.

Single agents plateau around ~10 tools and ~20 steps. Beyond that, multi-agent.

Five Architectures for Agentic Systems

Different ways to wire perception · planning · memory · execution · reflection (Bandi et al. 2025)

REACT

Single agent



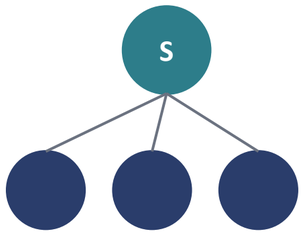
Linear loop: perceive ⟶ reason ⟶ act ⟶ observe.

BEST FOR

Bounded tasks · rapid prototyping

SUPERVISOR

Hierarchical



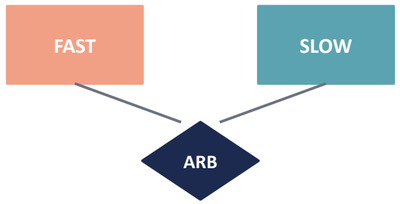
Central orchestrator delegates to specialized workers.

BEST FOR

Decomposable multi-stage tasks

HYBRID R-D

Reactive + Deliberative



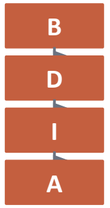
Fast loop for events + slow loop for planning, mediated by arbitrator.

BEST FOR

Real-time ops with long horizon

BDI

Belief-Desire-Intention



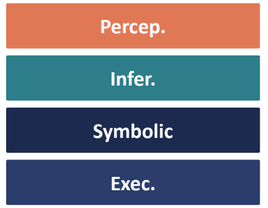
Explicit mental states: beliefs ⟶ desires ⟶ intentions ⟶ action.

BEST FOR

Explainable safety-critical workflows

NEURO-SYMBOLIC

Layered



Neural perception + probabilistic inference + symbolic planner.

BEST FOR

Uncertainty + transparent audit

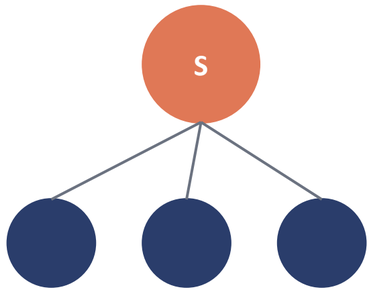
Choose by control flow, not vocabulary. Most production systems mix two patterns: a Supervisor + ReAct workers is the dominant 2026 blueprint.

Source: Bandi et al. (2025), Future Internet 17(9):404, Figures 5-9

Multi-Agent Topologies

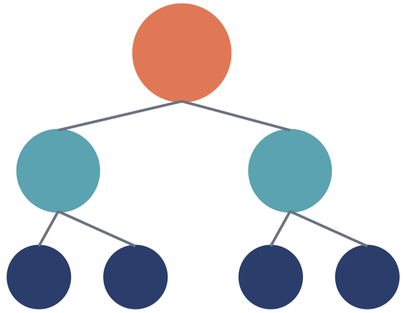
Four patterns dominate production deployments in 2026

SUPERVISOR



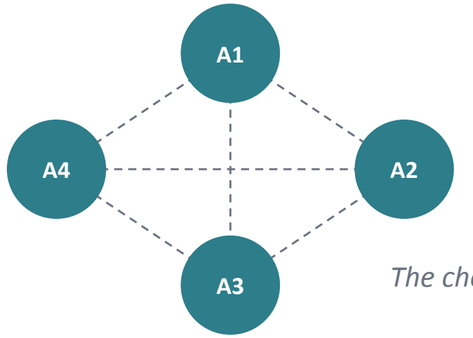
Central orchestrator delegates to specialized workers

HIERARCHICAL



Supervisors of supervisors — enterprise scale

NETWORK / SWARM



Peer agents with handoffs (OpenAI Agents SDK style)

SEQUENTIAL PIPELINE



Agents in chain, each handing off output to the next

The choice depends on coordination overhead vs parallelism: tight coupling → supervisor; loose coupling → swarm.

Why Multi-Agent? Three Structural Benefits

And one structural cost you must accept

①

Specialization

One agent = one role + its own tools, prompts, model. Higher accuracy than a generalist holding too many concerns.

②

Parallelism

Workers explore independent paths simultaneously. Total time drops; throughput rises on breadth-first tasks.

③

Context Isolation

Each agent has its own context window. No pollution from unrelated information. Cleaner reasoning per worker.

THE COST · Multi-agent can burn ~15× the tokens of a single chat. Deploy where unit value justifies it.

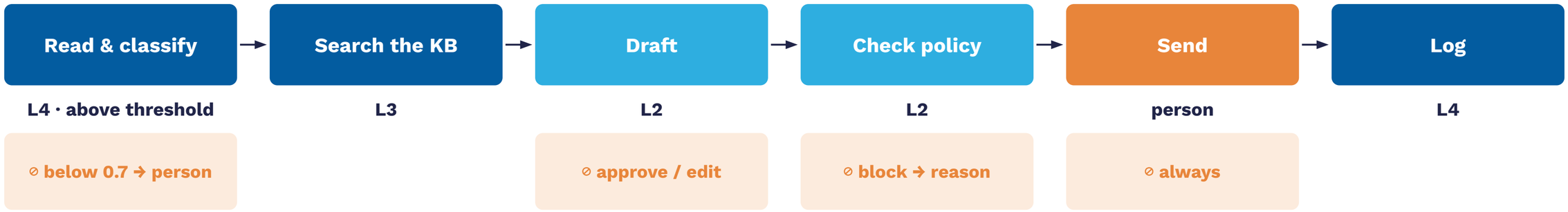
DECODE THE ARCHITECTURE

Workflow or agent? The question the exam and the CFO ask

	Workflow	Agent
Who decides the path	you, in the spec — a drawing	the model, at run time — a loop
Cost per ticket	predictable: the same steps every time	variable: as many model calls as the loop takes
When it fails	at a known step, with a known fallback	anywhere in the loop; errors compound
Can you explain it	yes: “step 3 returned no article”	hard: “it decided to call the refund tool”
Trust level it can carry	L2–L4 per step, chosen by you	L4 by construction — unless you fence it
Use it when	the path can be drawn in advance	the path really depends on what it finds

Your support agent is a workflow: classify, retrieve, draft, check. Chevrolet's bot was an agent with a price tool and no fence. The exam will show you a product and ask which one it is — and why.

Human checkpoints and the override switch — the Chevrolet answer



⊘ A HUMAN CHECKPOINT IS

- a place in the flow where a named person **approves, corrects or takes over** — set by the design, not by the model's confidence
- **two kinds:** approval (“send”) and abstention (the system withdraws: payroll, a complaint, a medical situation)
- each one has an owner and a service level: who, and within how long

□ THE OVERRIDE SWITCH IS

- **one action that drops the whole system to L1** — suggestions only, nothing sent, nothing paid
- a feature flag with an owner, tested before launch
- Chevrolet's switch was “unplug the bot”. That is not a switch. And there was no checkpoint before a price was stated — a rung-1 rule would have done it.

Exercise 1: decode a described system — 8 min, pairs

AN AIRLINE ASSISTANT FOR 2,000 CONTACT-CENTRE AGENTS · ILLUSTRATIVE

- When a call starts, the assistant pulls the passenger's booking and recent interactions and listens to the live transcript.
- It shows the agent three things: a summary of the situation, the policy paragraphs that apply (rebooking, compensation, baggage), and a suggested next action with a one-line script.
- For compensation claims it pre-fills the form with the amount the policy allows; the agent reviews and submits.
- If the transcript mentions a medical situation or a complaint about staff, the assistant stops suggesting and flags a supervisor.
- Every suggestion is logged with the passage it came from. Agents can switch the assistant off for a call with one click.

ON YOUR SHEET

- the components, in order
- the pattern of each one
- the trust level per step
- the human checkpoints — and their kind
- the override switch
- one metric, one main risk
- **workflow or agent? why?**

Exercise 1 — a reference decoding

Component	Pattern	Trust level	Checkpoint
Transcription → summary	prompt chaining	L2 — shown to the agent	—
Policy retrieval	retrieval (augmented LLM)	L2	—
Next-action suggestion + script	chaining, grounded on the passages	L2	the agent decides
Compensation form pre-fill	rule-based: the policy amount	L2	⊘ approval: review and submit
Medical / complaint flag	routing on a rule	L1 — the assistant withdraws	⊘ abstention: supervisor
Log with the source passage	rule-based	L4	—

A workflow, not an agent: fixed steps, gates between them, no tool loop. The switch: one click, per call, by the agent. Metric: handling time at constant quality. Main risk: a wrong amount pre-filled and submitted without a real review — and a missed medical flag.

WORKSHOP · 40 MIN · TEAMS

Build the hybrid: four steps, four commits, one number

1 · Router · 10 min

TF-IDF + logistic regression on tickets_synthetic.csv → category + confidence. Commit: step-1-router.

2 · Routing rule · 5 min

A threshold: below it, the ticket goes to a person. Choose it against the cost of a wrong route (payroll routed as “how do I” is expensive). Commit: step-2-routing-rule.

3 · Drafter · 15 min

Retrieval over the three KB articles (provided) + a prompt that cites its passages. No API key? Use the mock drafter — the architecture is the same. Commit: step-3-drafter.

4 · Policy checker + eval · 10 min

Rules: no price, no refund, no legal advice. Then run evals/tickets_v0.csv and read your number. Commit: step-4-policy, eval-v0.

THE RULES

- **Starter repo: support-copilot-starter.zip**
- One commit per step, one person per commit
- Never tune while looking at evals/ — that is drill #2
- Step 4 may finish at home
- Teams whose agent is not a support copilot: same skeleton, your own 20 tickets

One ticket through the four steps — and what each step decides

Ticket E01 · “How do I export last month's payroll run as a CSV for our accountant?” — from evals/tickets_v0.csv, 25 cases, labelled before the run

	WHAT THE STEP DOES	WHAT COMES OUT FOR E01	THE PRODUCT DECISION
1 · Route **rung 2 · classical ML**	TF-IDF + logistic regression, trained on 200 synthetic tickets	**how_do_i · confidence 0.83**	**you decide:** the categories, and the training data
2 · Routing rule **rung 1 · a rule**	draft only if how_do_i AND confidence ≥ 0.70 AND no money word (refund, price, invoice…)	**→ draft**	**you decide:** the threshold — against the cost of a wrong route — and the human-only list
3 · Retrieve + draft **rung 3 · pretrained model**	top-3 chunks of the KB by heading, score ≥ 0.05 · a prompt that uses only those chunks and cites them · nothing found → NO_DRAFT	**“Open Payroll → Runs, select the month… [KB-214 · Export a run as CSV]”**	**you decide:** the sources it may read, the citation format, the fallback
4 · Policy check **rung 1 · a rule**	block if a price, “refund”, “legal”… · a blocked draft is shown with its reason, never approvable	**pass → Karim approves, edits or escalates (L2)**	**you decide:** what the model is never trusted with

Run on 25 cases: 8/20 how-do-I sendable as is = 40 % (criterion 45 %) · 0 drafts on human-only · 9 misses have one cause: the KB has no “Release notes 4.12” — a line for spec v2, not a prompt tweak.

WORKSHOP · 40 MIN · YOUR TEAM, YOUR AGENT

Your agent in four steps: draw it, decide it, write the first five cases

1 · Draw the pipeline · 8 min

- What comes in, what goes out — one line
- The steps in order; one rung per step (1 rule · 2 classical ML · 3 pretrained · 4 custom)
- Mark every human checkpoint ⊘ and the switch □

2 · The routing rule · 7 min

- What is human-only, by name (money, grades, a P1, a complaint…)
- The threshold below which a person takes over
- The cost of a wrong route, both ways — which error is expensive?

3 · The drafter or decider · 10 min

- What it may read (your KB, policy, data) — and what it may not
- What it must cite or show as evidence
- Nothing found / low confidence / API down → what happens

4 · Policy check + five cases · 10 min

- Three things the model is never trusted with
- Five cases in evals/eval_set_v0.csv: easy · hard · out of scope · another segment · your trap
- What number you would report on 6 November, against your criterion

YOU LEAVE WITH

- **04_architecture.md v0** — the drawing (a photo of the whiteboard is fine), the four decisions
- **evals/eval_set_v0.csv** — five labelled cases
- **governance.md · S4** — your checkpoints, who owns the switch

35 min, then 5 min: swap with the next team — they read your drawing and say where the money is and where the person is. If they cannot, redraw.

At home, before 6 November: run the Paylio starter once (support-copilot-starter.zip, four commits, 30 min) — then swap in your own cases and documents. The four files you touch are the four decisions you just made.

Before you leave — and before 6 November

LEAVE THE ROOM WITH

- **/prototype —** the four steps, four commits — step 4 may be unfinished
- **evals/report_v0.md —** your number against your criterion, and the three most frequent misses
- **governance.md · Session 4 —** your human checkpoints and who owns the override switch

BEFORE 6 NOVEMBER

- [ ] **04_architecture.md —** the diagram, the component table, patterns, trust levels, checkpoints ⊘, switch □
- [ ] **spec.md v2 —** every eval miss becomes a line
- [ ] **evals/eval_set_v0.csv → 50 cases —** on the way to 200 by Session 5
- [ ] **03 homework, if not done —** brief v2 after the three conversations, the Lean Canvas

NEXT · S5 · 6 NOVEMBER

- Metrics: from your criterion to three business metrics, with expected effects
- The eval-set grows to 200 — and becomes a metric you can trust
- The reviews.csv data audit, on real data



SESSION 4

References & going further

METHODS

- Anthropic, Building effective agents (December 2024) — the augmented LLM, five workflow patterns, workflows vs agents
- A. Karpathy, “vibe coding” (February 2025); M. Fowler, Vibe Coding, martinfowler.com/bliki (2025)
- Zinkevich, Rules of Machine Learning (Google) — rule #1: launch without ML if you can
- Streamlit and FastAPI documentation — the demo surface and the production surface
- Session 3: Brynjolfsson & Mitchell (2017) — automate, augment or keep human

CASES CITED TODAY

- AI Incident Database, incident 622 — Chevrolet of Watsonville (18 December 2023)
- Gartner press release (25 June 2025) — over 40 % of agentic AI projects cancelled by end 2027; “agent washing”
- Klarna press release (27 Feb 2024); Bloomberg (8 May 2025) — the escape path, added late
- The airline assistant of exercise 1 and the Paylio prototype are illustrative, built for the course

The spec is the decision. The eval-set is the judge. The switch has an owner.



01

A prototype in a day — to find out which assumptions were wrong

02

One rung per part; demo data and evaluation data never meet

03

Workflow when you can draw the path; an agent only when you cannot

04

Every checkpoint has an owner; the off switch is one action, tested

Bring 04_architecture.md and your eval number on 6 November